

1. 设独立同分布的随机变量 $X_1, \dots, X_n$ 取自伯努利分布总体 $Bernoulli(\theta)$, 且 $0 \leq \theta \leq \frac{1}{2}$

(a) 求 θ 的矩估计量和最大似然估计量;

(b) 求以上两种估计量的均方误差。

$$(a) E(X) = \theta \quad \hat{\theta}_{ML} = \bar{X}$$

$$P(\bar{X}) = \theta^{\sum_{i=1}^n x_i} (1-\theta)^{n-\sum_{i=1}^n x_i}$$

$$l(\theta) = \sum_{i=1}^n x_i \ln \theta + (n - \sum_{i=1}^n x_i) \ln(1-\theta)$$

$$\frac{\partial l}{\partial \theta} = 0 \Rightarrow \frac{n\bar{X}}{\theta} - \frac{n-n\bar{X}}{1-\theta} = 0 \quad n\bar{X} - \theta n\bar{X} - n\theta + n\bar{X}\theta = 0$$

$$\hat{\theta}_{MLE} = \min\left(\frac{1}{2}, \bar{X}\right)$$

$$(b) E(\hat{\theta}_{ML}) = \theta \quad \text{Var}(\hat{\theta}_{ML}) = \frac{\theta(1-\theta)}{n}$$

$$MS_E(\hat{\theta}_{ML}) = \text{Var}(\hat{\theta}_{ML}) + (\text{Bias } \hat{\theta}_{ML})^2 = \frac{\theta(1-\theta)}{n}$$

$$\text{若 } \bar{X} \leq \frac{1}{2},$$

$$\text{Var}(\hat{\theta}_{MLE}) = \frac{\theta(1-\theta)}{n} \quad E(\hat{\theta}_{MLE}) = \theta$$

$$MS_E(\hat{\theta}_{MLE}) = \frac{\theta(1-\theta)}{n}$$

$$\text{若 } \bar{X} > \frac{1}{2}$$

$$E(\hat{\theta}_{MLE}) = \frac{1}{2}, \quad \text{Var}(\hat{\theta}_{MLE}) = 0$$

$$MS_E(\hat{\theta}_{MLE}) = \left(\frac{1}{2} - \theta\right)^2$$

2. 设独立同分布的随机变量 $X_1, \dots, X_n$ 取自正态总体 $N(\mu, \sigma^2)$ 分布

(a) 求统计量 $T = \sum_{i=1}^n \alpha_i X_i$ 是 μ 的无偏估计量所需要满足的条件;

(b) 在这些线性无偏估计量中, 哪个具有最小的方差? 请求出最小的方差并给出取得条件。

$$(a) E(T) = \left(\sum_{i=1}^n \alpha_i \right) \mu$$

$$E(T) = \mu \Leftrightarrow \left(\sum_{i=1}^n \alpha_i \right) = 1$$

$$(b) \text{Var}(T) = \sum_{i=1}^n \alpha_i^2 \sigma^2$$

$$\arg \min_{\substack{\alpha_1, \dots, \alpha_n \\ \alpha_1 + \dots + \alpha_n = 1}} \sum_{i=1}^n \alpha_i^2 \sigma^2 \Rightarrow \alpha_1 = \dots = \alpha_n = \frac{1}{n}$$